

Smart PV Controller



Active Safety

AI Powered Arcing Protection



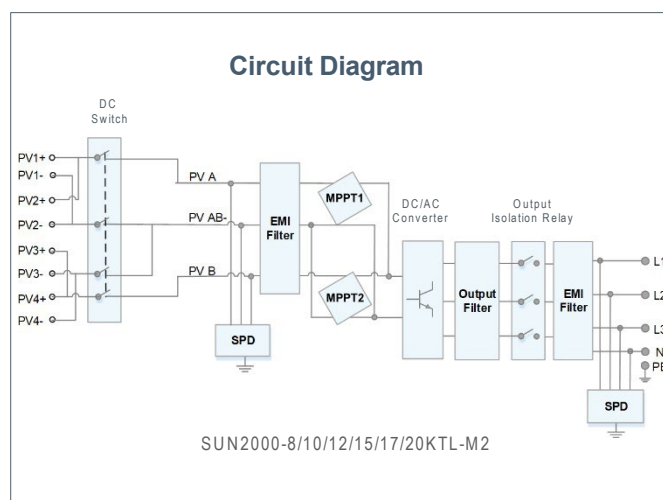
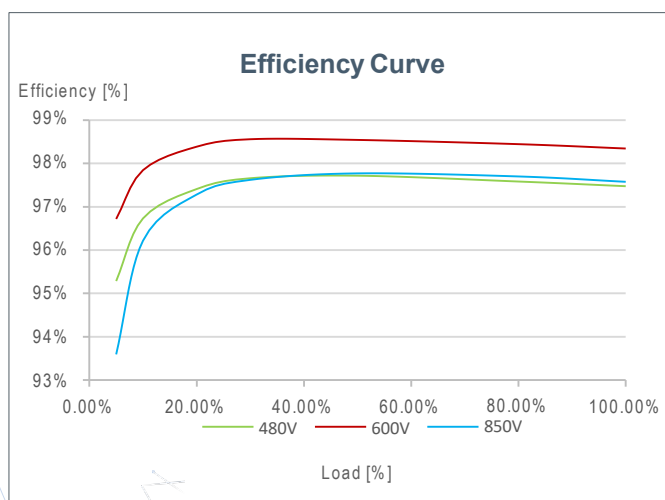
Higher Yields

Up to 30% More Energy with Optimizer



Flexible Communication

WLAN, Fast Ethernet, 4G
Communication Supported



Technical Specification

Technical Specification	SUN2000 -8KTL-M2	SUN2000 -10KTL-M2	SUN2000 -12KTL-M2	SUN2000 -15KTL-M2	SUN2000 -17KTL-M2	SUN2000 -20KTL-M2
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Efficiency

Max. efficiency	98.50%	98.50%	98.50%	98.65%	98.65%	98.65%
European weighted efficiency	97.80%	98.00%	98.00%	98.30%	98.30%	98.30%

Input

Recommended max. PV power ¹	12,000 Wp	15,000 Wp	18,000 Wp	22,500 Wp	25,500 Wp	30,000 Wp
Max. input voltage ²	1,080 V					
Operating voltage range ³	160 V ~ 950 V					
Start-up voltage	200 V					
Rated input voltage	600 V					
Max. input current per MPPT	27 A ⁴					
Max. short-circuit current	39 A					
Number of MPP trackers	2					
Max. number of inputs	4					

Output

Grid connection	Three phase					
Rated output power	8,000 W	10,000 W	12,000 W	15,000 W	17,000 W	20,000 W
Rated AC Apparent power	8,000 VA	10,000 VA	12,000 VA	15,000 VA	17,000 VA	20,000 VA
Max. apparent power	8,800 VA	11,000 VA	13,200 VA	16,500 VA	18,700 VA	22,000 VA
Rated output voltage	230 Vac / 400 Vac, 3W + N + PE					
Rated AC grid frequency	50 Hz / 60 Hz					
Max. output current	13.4 A	17 A	20 A	25.2 A	28.5 A	33.5 A
Adjustable power factor	0.8 leading ... 0.8 lagging					
Max. total harmonic distortion	≤ 3 %					

Features & Protections

Input-side disconnection device	Yes
Anti-islanding protection	Yes
AC over-current protection	Yes
AC short-circuit protection	Yes
AC over-voltage protection	Yes
DC reverse-polarity protection	Yes
DC surge protection	TYPE II
AC surge protection	Yes, compatible with TYPE II protection class according to EN/IEC 61643-11
Residual current monitoring unit	Yes
Arc fault protection	Yes
Ripple receiver control	Yes
Integrated PID recovery ⁵	Yes

General Data

Operation temperature range	-25 ~ + 60 °C (-13 °F ~ 140 °F)
Relative humidity	0 % RH ~ 100% RH
Max. operating altitude	0 ~ 4,000 m (13,123 ft.) (Derating above 2000 m)
Cooling	Natural Convection
Display	LED Indicators; Integrated WLAN + FusionSolar App
Communication	RS485; WLAN/Ethernet via Smart Dongle-WLAN-FE (Optional) 4G / 3G / 2G via Smart Dongle-4G (Optional)
Weight (with mounting plate)	25 kg
Dimensions (W x H x D) (incl. mounting plate)	525 x 470 x 262 mm (20.7 x 18.5 x 10.3 inch)
Degree of protection	IP65
Country of Manufacture	China

Optimizer Compatibility

DC MBUS compatible optimizer	SUN2000-450W-P
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Standard Compliance (more available upon request)

Safety	EN/IEC 62109-1, EN/IEC 62109-2
Grid connection standards	G98, G99, EN 50549, CEI 0-21, CEI 0-16, VDE-AR-N-4105, VDE-AR-N-4110, AS/NZS 4777.2 2020, C10/11, ABNT, VFR 2019, RD 1699, RD 661, PO 12.3, TOR D4, IEC61727, IEC62116, DEWA

*1 Inverter max input PV power is 40,000 Wp when long strings are designed and fully connected with SUN2000-450W-P power optimizers.

*2 The maximum input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage inverter.

*3 Any DC input voltage beyond the operating voltage range may result in inverter improper operating.

*4 The MPPT voltage of each PV string must exceed the lower limit of Full Power MPPT Voltage Range. (Full Power MPPT Voltage Range: 12KTL@360~850V, 15KTL@380~850V, 17KTL@400~850V, 20KTL@450~850V)

*5 SUN2000-8-20KTL-M2 raises potential between PV- and ground to above zero through integrated PID recovery function to recover module degradation from PID. Supported module types include: P-type (mono, poly)